Supplementary Material for "M^{3}UCD: A Multi-task Multimodal Metaphor Understanding Challenge Dataset for LLMs"

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Comprehensive annotation guidelines

Metaphor Occurrence Annotation

(I) True: The "True" category encompasses content that incorporates metaphorical language or imagery, meaning that one concept is described in terms of another to create a parallel that enhances understanding, evokes emotions, or illustrates abstract ideas. This category includes: (1) Explicit Metaphors: Direct comparisons where one thing is stated to be another. (2) Implicit Metaphors: Comparisons that are suggested rather than directly articulated. (3) Analytical Metaphors: Metaphors used in analytical contexts to explain complex ideas. Metaphor enhances expression by providing emotional resonance and engaging the audience's imagination, prompting them to explore the underlying meanings and connections that the metaphor implies.

(II) False: The "False" category consists of content that does not use metaphorical language or imagery and presents information in a literal, straightforward manner. It is mainly manifested as clear and direct multimodal expression that communicates objective information without any metaphorical elements.

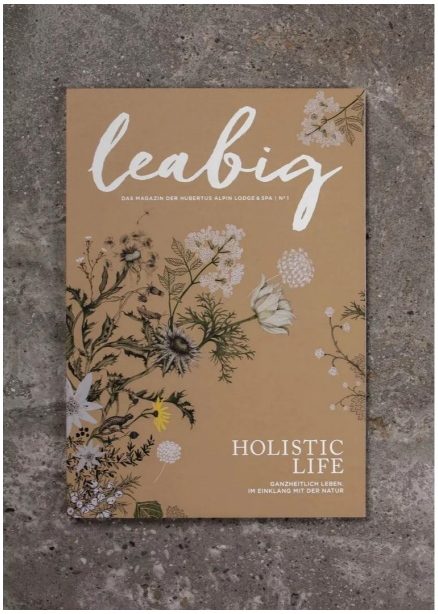
The metaphor occurrence annotation instances are illustrated in Figure 1.

Chinese: 愿每一块干旱的土地都能平安渡过。
English: May every arid land be able to pass through safely.



Metaphor: True

Chinese: 冬日暖阳，一本书，一杯茶，陪你一起度过一下午的惬意时光。
English: Bask in the warm winter sun, with a book and a cup of tea, spending a leisurely afternoon in blissful company.



Metaphor: False

Figure 1: The metaphor occurrence annotation instances.

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Metaphor Categories, Domain & Modality Annotation

(I) Text: "Text" represents the textually dominant metaphor. The category refers to instances where the metaphorical meaning primarily resides in the textual content. In this case, the language used conveys the metaphor without significant reliance on the accompanying imagery. The text provides the main metaphorical insights and emotional resonance, while any images present serve to support or illustrate the text rather than being the primary source of meaning.

(II) Image: "Image" represents the visually dominant metaphor. The category pertains to instances where the metaphorical meaning is primarily conveyed through images rather than text. In this case, the visual elements play a critical role in establishing the metaphor, while the textual content may provide minimal or no metaphorical context. The imagery serves as the main vehicle for metaphorical expression, creating a strong visual impact.

(III) Complementary: Content classified as "Complementary" emphasizes the collaborative nature of text and imagery in conveying metaphorical meaning. In this case, the text and images work synergistically to enhance the metaphor, providing a richer and more nuanced understanding that neither could achieve alone. Both elements are integral in conveying the metaphor.

The metaphor categories, domain & modality annotation instances are illustrated in Figure 2.

Chinese: 炎炎夏日；忘记炎热忘记烦恼；嗨翻!
English: Sizzling summer days—shake off the heat and worries—let's crank up the fun!



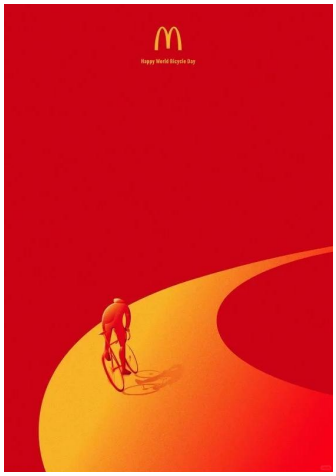
Source Modality: Image
Source Domain: 海浪/Ocean waves
Target Modality: Image
Target Domain: 橙汁/Orange juice
Metaphor Categories: Image

Chinese: 六一儿童节，童年心未泯，心怀世界梦；花样年华的童年，天真烂漫的故事。
English: Children's Day: Where youthful hearts beat with global dreams—childhood in the blossom of youth, painted with innocent stories.



Source Modality: Text
Source Domain: 花样年华/blossom age
Target Modality: Text
Target Domain: 童年/Childhood
Metaphor Categories: Text

Chinese: 麦当劳，世界自行车日快乐。
English: M. Happy world Bicycle Day.



Source Modality: Complementary
Source Domain: 自行车跑道/Bicycle track
Target Modality: Complementary
Target Domain: 麦当劳Logo/MacDonald logo
Metaphor Categories: Complementary

Figure 2: The metaphor categories, domain & modality annotation instances.

Emotion Annotation

(I) Happiness: Happiness encompasses positive sentiments characterized by feelings of joy, contentment, and a sense of well-being. It can manifest in various forms including relaxation, peace, and a striving for strength. Happiness is a fundamental emotion that reflects an individual's satisfaction and positive engagement with the world. It often leads to increased social interactions and enhances overall mental health.

(II) Love: Love is a profound emotional state that encompasses affection, deep care, and attachment towards someone or something. It includes romantic love, familial love, platonic love and so on. Love promotes bonding and connection between individuals. It plays a crucial role in interpersonal relationships and contributes to personal happiness and fulfillment.

(III) Anger: Anger is a strong emotional response characterized by feelings of frustration, annoyance, or hostility. It often arises in response to perceived injustices or obstacles. Anger can motivate individuals to confront challenges or injustices. However, if uncontrolled, it can lead to destructive behavior and conflict.

(IV) Sorrow: Sorrow is a deep feeling of sadness or grief often related to loss, disappointment, or despair. It can manifest as a quiet reflection or deep mourning. Sorrow can be a powerful response to life's challenges and losses. It often leads to introspection and can ultimately facilitate healing and growth over time.

(V) Fear: Fear is an emotional response to perceived threats or danger, characterized by feelings of anxiety, apprehension, or dread. Fear serves as a protective mechanism, alerting individuals to potential dangers and prompting them to take action to ensure safety.

(VI) Hate: Hate is an intense emotional response characterized by feelings of aversion, hostility, or extreme dislike towards someone or something. Hate can arise from fear, anger, or resentment. While it can motivate change, unchecked hate can lead to destructive conflict and societal division.

(VII) Surprise: Surprise is an emotional reaction to unexpected events or information, which can be either positive or negative, characterized by a sense of astonishment or disbelief. Surprise can enhance learning and adaptability by prompting individuals to pay attention to new information or changes in their environment.

The emotion annotation instances are illustrated in Figure 3.

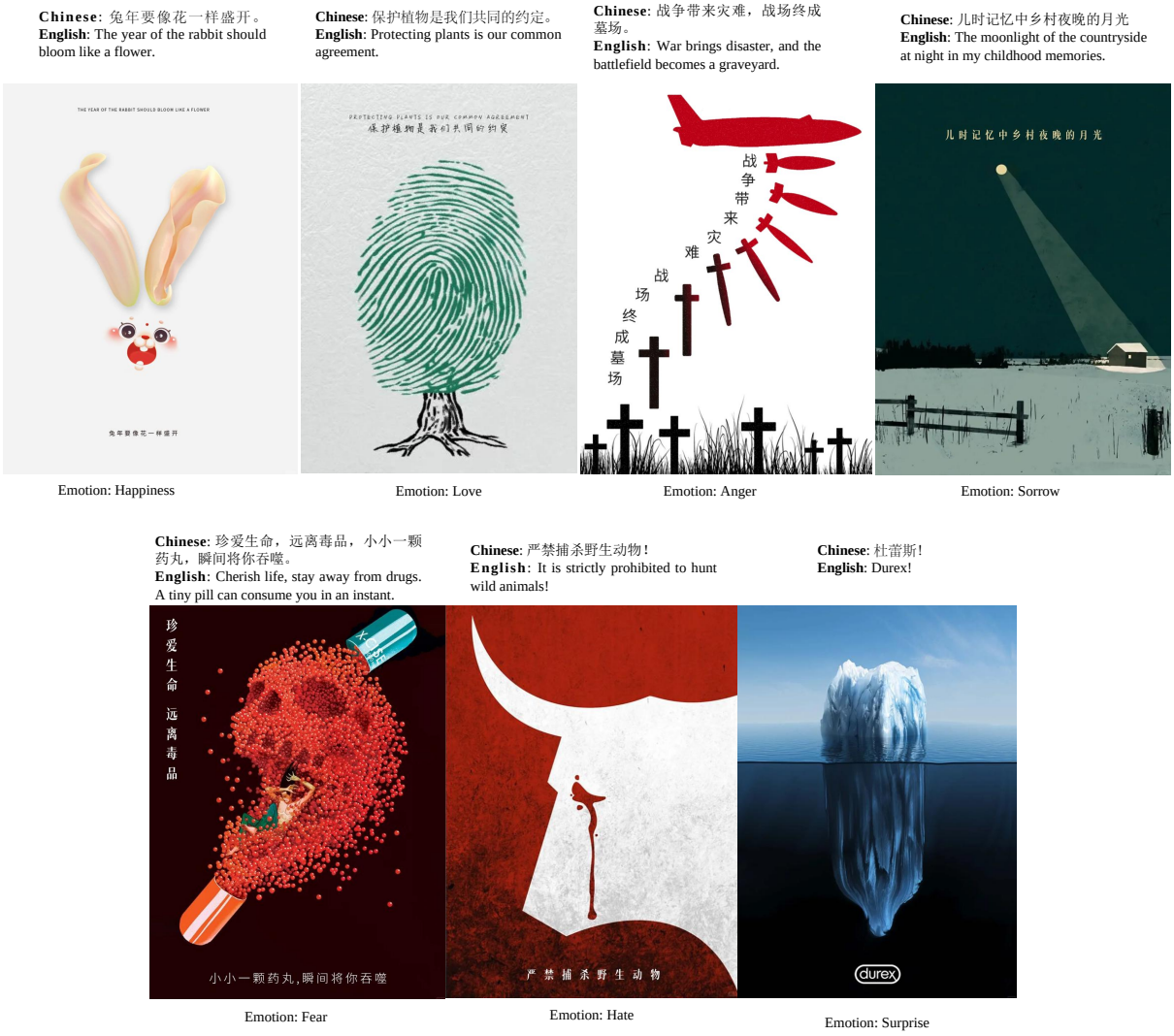


Figure 3: The emotion annotation instances.

Offensiveness Annotation

(I) No_Offensive: No_offensive content has no intention to attack, satirize, disparage, or offend any individual or group. It maintains a neutral or positive tone and promotes respectful communication. It is often educational or constructive, promoting a peaceful exchange of ideas without provoking any negative responses.

(II) Slightly Offensive: Slightly offensive content includes mild remarks or implications that may cause discomfort or irritation but are not intended to cause harm. It may involve light sarcasm or trivial negative comments. It can serve as an icebreaker in conversations, although it should be used cautiously to avoid misinterpretation.

(III) Moderately Offensive: Moderately offensive content contains direct disparaging remarks, insults, or expressions of intolerance that can seriously offend or alienate individuals or groups. It often reflects negative attitudes or

stereotypes. This level of offensiveness often escalates discussions into arguments and can alienate individuals, creating a hostile environment. It's crucial to approach this type of content with sensitivity and an understanding of its potential impact.

The offensiveness annotation instances are illustrated in Figure 4.

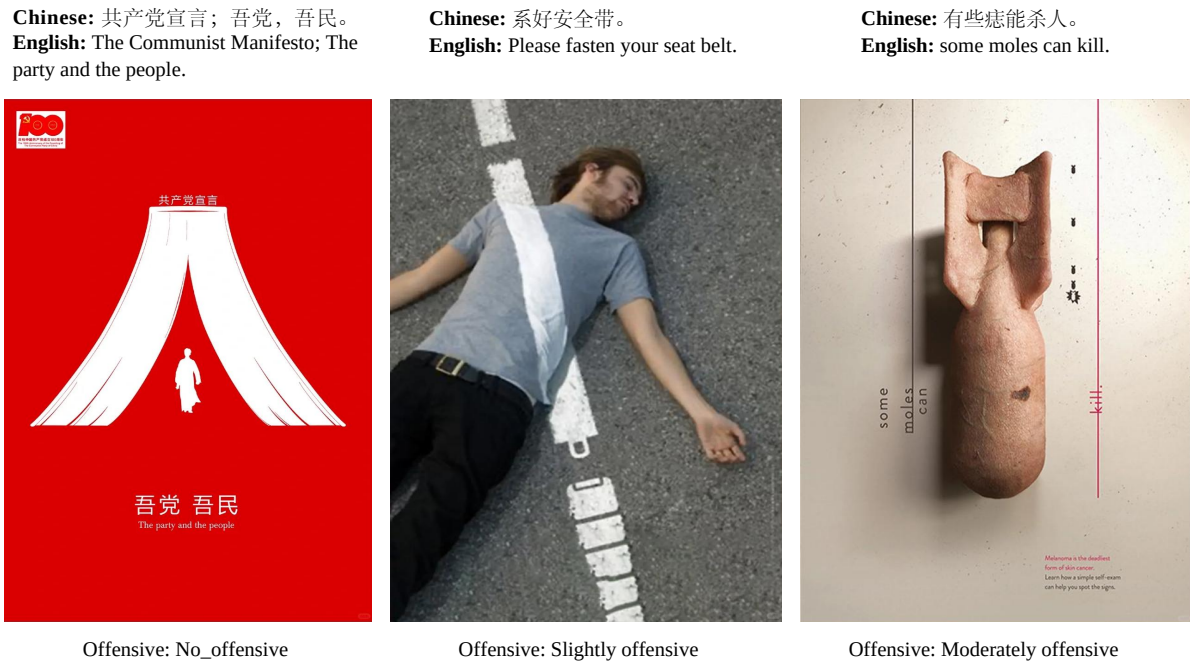


Figure 4: The offensiveness annotation instances.

Intention Annotation

(I) Interactive: Interactive intention emphasizes fostering engagement and communication between two or more parties. It aims to encourage responses, feedback, or participation from the audience. This type of intention is vital in social media, marketing, and community engagement where feedback and participation are essential.

(II) Expressive: Expressive intention serves to communicate inner thoughts and feelings, promoting empathy and understanding among audiences. It can strengthen connections through shared experiences and emotional resonance.

(III) Entertaining: Entertaining intention aims to amuse or engage the audience through humor, creativity, or enjoyable content. It focuses on capturing attention and providing enjoyment. It is often used in social media, advertisements, and creative content to engage audiences and enhance their experience.

(IV) Offensive: Offensive intention involves communication that is intended to insult, demean, or provoke negative reactions from individuals or groups. It often reflects hostility or intolerance. Recognizing this intent is crucial for moderating discussions and maintaining respectful communication.

(V) Other: The "Other" category encompasses intents that do not clearly fit into the previous categories and allows for flexibility in labeling content that does not conform to established intents. This may include content that is ambiguous, context-specific, or lacks a distinct intent. It acknowledges the complexity of communication and the nuances that may arise in different contexts.

The intention annotation instances are illustrated in Figure 5.

Humor Annotation

(I) True: The "True" category encompasses content that successfully elicits laughter or amusement from the audience. This includes positive humor (which uplifts or entertains) and negative humor (which may involve sarcasm, irony, or dark humor). It also spans varying degrees of humor intensity, from lighthearted to more pronounced comedic expressions. Humor that falls into the "True" category serves to entertain, engage, or elicit emotional responses from the audience. Positive humor fosters connection and enjoyment, while negative humor can provide social commentary or critique. The effectiveness of humor often relies on timing, relatability, and context, making it a nuanced form of communication.



Figure 5: The intention annotation instances.

(II) False: The "False" category includes content that does not elicit humor or amusement and may be perceived as serious, dull, or lacking in comedic elements. This includes content that fails to connect with the audience's sense of humor or comes across as inappropriate or offensive without any comedic intent.

The humor annotation instances are illustrated in Figure 6.

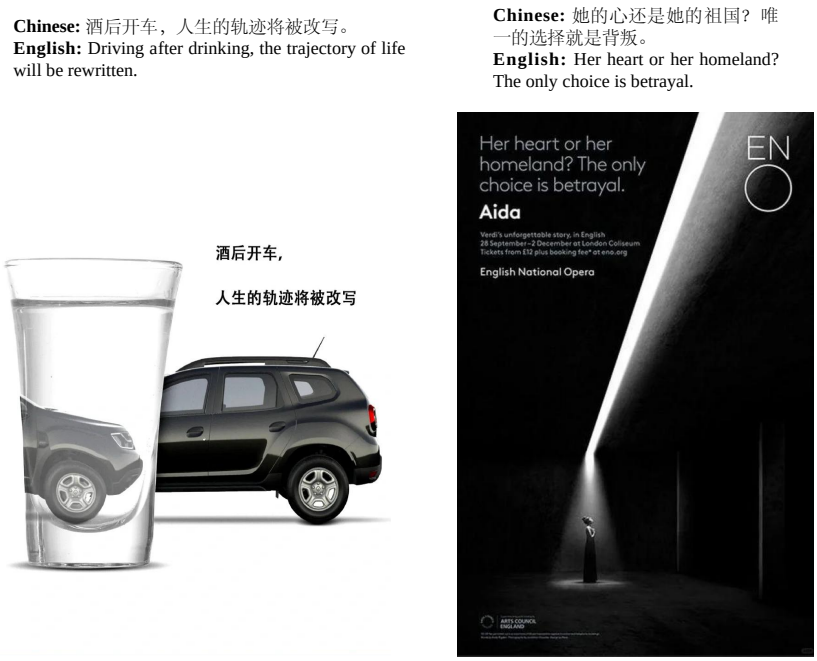


Figure 6: The humor annotation instances.

Sarcasm Annotation

(I) True: The "True" category encompasses content that effectively utilizes sarcasm to convey a meaning opposite to the literal interpretation. This includes both positive sarcasm (where the intent is to humorously uplift or praise in a roundabout way) and negative sarcasm (which may involve mockery, criticism, or disdain). Sarcasm in the "True" category serves to critique, humor, or express one's feelings in a subtle manner. It often relies on context, tone, and delivery to convey the underlying message, making it a powerful tool for social commentary or humor. The effectiveness of sarcasm hinges on the audience's ability to recognize the discrepancy between the literal and intended meanings.

(II) False: The "False" category includes content that does not exhibit sarcasm and is either straightforward or lacks any ironic undertones. This category may encompass sincere statements or neutral observations that are not intended

Chinese: “留给他们的时间不多了，而非塑料。”
English: "Time is running out for them, not for plastic."



Sarcasm: True

Chinese: 别怕，我罩着你，为你抵挡所有紫外线伤害，玫莉楚小白管防晒乳。
English: Don't be afraid. I'll protect you and shield you from all the ultraviolet rays. Morningskin White Tube Sunscreen Lotion.



Sarcasm: False

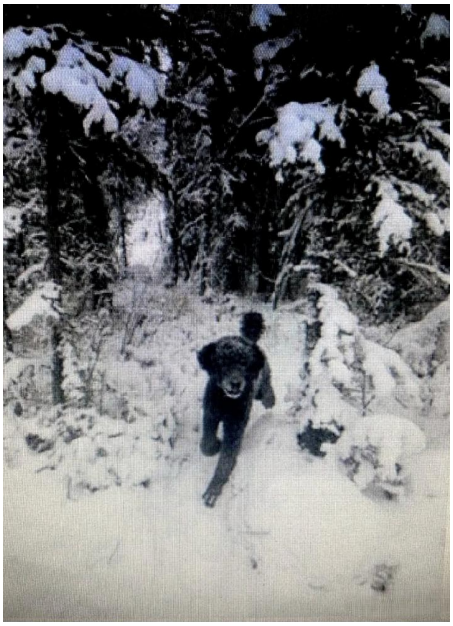
Figure 7: The sarcasm annotation instances.

Chinese: 要么是你的浪漫晚餐，要么是我的生命。
English: Your romantic dinner or my life.



Hyperbole: True

Chinese: 你到底看见了一个在奔跑的人，还是看见了一条奔跑的狗?
English: Do you see a man running, or do you see a dog running?



Hyperbole: False

Figure 8: The hyperbole annotation instances.

to mock or critique. This category emphasizes the contrast between straightforward communication and the nuanced nature of sarcasm.

The sarcasm annotation instances are illustrated in Figure 7.

Hyperbole Annotation

(I) True: The "True" category refers to content that employs hyperbole, which is a figure of expression that involves hyperbole for emphasis or effect. This can include statements that magnify or diminish characteristics well beyond factual accuracy, creating an impression that is not aligned with reality. It often seeks to engage the audience by exaggerating a point for rhetorical impact.

(II) False: The "False" category includes content that does not employ hyperbole and presents information in a straightforward and factual manner. This includes accurate representations of size, characteristics, or situations without any hyperbole. This category underscores the importance of distinguishing between factual information and creative, exaggerated expressions.

The hyperbole annotation instances are illustrated in Figure 8.

Type	*Model*	*Metaphor-rich Semantic Orientation Tasks*						*Metaphor-rich Pragmatic Figurativeness Tasks*						*Partial Metaphor Reasoning Tasks*							
		EA		*OD*		*IR*		*HumorD*		*SD*		*HyperD*		*MD*		*SDD*		*TDD*		*MC*	
		Prec.	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*	*Prec.*	*Rec.*
	Random	14.13	13.63	33.88	32.98	19.08	18.46	50.76	50.79	49.96	49.93	50.45	50.47	50.38	50.41	30.82	30.63	31.89	32.10	34.46	35.20
PTMs	Bi-LSTM	47.64	40.13	78.81	45.60	51.22	45.00	72.60	69.63	82.00	74.83	75.53	74.56	73.34	68.97	60.98	55.18	53.09	52.25	60.91	56.40
	BERT	44.47	43.90	69.30	56.80	55.51	50.11	75.57	75.63	81.40	80.24	78.81	77.13	76.87	77.12	64.94	59.93	58.94	57.53	65.63	63.21
	ResNet50	30.04	28.93	47.57	50.23	35.25	34.78	68.16	67.60	69.58	73.97	76.86	75.75	67.37	66.87	51.06	47.63	45.58	44.99	48.52	48.24
	ViT	28.04	21.45	53.13	40.66	34.07	29.80	70.82	66.84	68.59	68.79	73.74	74.50	68.47	68.31	52.60	45.08	43.85	43.48	53.77	47.53
	MAE	26.75	25.91	58.12	43.63	32.95	30.44	70.24	65.31	67.52	66.19	72.71	72.88	68.17	65.04	50.30	48.27	45.09	43.50	55.63	42.93
	DeiT	33.07	29.87	56.12	49.30	37.54	34.76	70.13	70.47	76.58	65.55	74.13	75.08	70.24	68.65	50.75	48.24	48.68	47.77	55.80	47.97
	SwinT	32.49	26.23	55.76	45.92	36.66	33.16	70.72	70.24	71.86	74.92	75.24	75.76	69.82	71.21	55.61	49.60	49.66	44.75	56.72	48.95
	CLIP	46.56	45.60	64.54	58.56	53.97	54.55	75.97	75.98	83.05	82.51	78.83	78.53	75.55	75.07	64.90	61.96	59.05	59.22	63.90	62.51
	FLAVA	23.33	23.36	46.16	42.39	32.83	30.83	66.38	66.15	73.82	67.08	69.72	67.88	67.41	66.40	44.43	39.35	47.72	43.99	19.85	33.33
LLMs	ChatGLM3-6B-Base	25.66	16.06	37.90	35.56	18.44	22.56	57.04	54.62	53.07	54.91	51.43	51.18	53.49	51.21	30.13	33.61	28.09	33.19	33.75	33.72
	LLaMA3-8B-Instruct	21.54	22.45	41.16	40.04	19.68	20.46	57.64	55.82	54.77	57.87	52.10	52.10	53.79	52.47	35.60	32.45	20.37	31.24	36.62	33.62
	InternLM2.5-7B-Chat	31.83	24.99	49.28	41.71	26.32	22.59	57.08	57.32	56.47	59.26	53.49	53.63	54.56	54.27	33.46	33.00	35.21	33.51	44.48	33.29
	LLaMA3.1-8B-Instruct	21.09	15.54	34.43	34.15	7.79	19.33	53.62	50.79	49.54	49.84	50.26	50.06	50.56	50.13	35.17	33.76	31.39	33.38	32.56	34.41
	MiniCPM3-4B-Chat	27.95	25.02	45.77	46.22	32.96	27.31	54.86	53.88	62.63	59.42	57.27	54.49	54.86	54.99	20.02	33.59	28.96	33.55	37.86	33.33
	GPT-3.5-turbo	32.31	32.60	39.65	42.85	32.18	24.92	61.33	56.95	53.01	54.40	45.55	45.76	57.38	57.60	20.71	33.18	35.47	33.70	38.43	33.39
MLLMs	InternVL2-1B	26.87	26.36	39.15	35.30	13.31	20.28	37.62	46.46	40.68	49.98	20.13	50.00	42.38	46.30	29.92	33.24	15.10	33.17	49.17	33.61
	InternVL2-2B	32.56	27.32	43.96	40.69	27.83	28.83	54.11	52.73	70.09	54.50	46.02	46.36	53.33	53.57	34.51	35.30	41.80	33.37	29.76	34.69
	InternVL2-4B	28.84	29.74	50.06	43.95	28.58	26.88	60.61	60.81	54.55	51.65	54.40	52.51	69.11	50.47	27.74	34.26	22.18	32.36	29.61	33.63
	LLaVA1.5-7B-Chat	17.66	19.01	36.62	38.24	18.04	21.50	50.72	50.14	51.28	51.64	50.50	50.50	48.24	49.47	31.09	32.17	26.25	34.53	31.97	32.95
	Qwen2-VL-7B-Instruct	27.17	28.42	46.14	43.72	32.18	22.72	58.94	57.87	59.42	59.07	55.26	53.35	53.34	53.56	36.52	35.49	37.21	37.74	35.95	35.58
	InternVL2-8B	42.08	32.87	46.39	33.62	29.68	31.73	64.40	63.98	67.65	75.12	57.04	57.28	58.09	53.77	33.91	35.06	36.48	33.68	36.94	35.09
	Llama-3.2-11B-Vision-Instruct	25.69	26.09	47.04	48.74	30.98	25.90	57.45	56.22	57.90	62.29	57.38	57.08	58.52	50.15	32.69	32.67	34.57	34.54	35.06	33.43
	InternVL2-26B	36.32	36.19	59.55	36.00	31.41	33.55	69.40	67.32	68.52	76.06	65.88	66.18	71.29	55.10	43.07	35.84	33.39	39.49	35.00	34.76
	InternVL2-40B	36.51	36.35	49.41	42.22	34.69	28.91	69.67	70.15	67.48	78.38	58.90	59.10	70.87	54.56	34.07	37.94	35.15	39.24	38.32	37.92
	Qwen2-VL-72B-Instruct	30.58	30.72	52.93	40.24	24.51	25.72	66.96	67.36	73.48	57.43	62.11	60.06	68.00	56.76	38.23	37.79	37.57	37.94	44.36	33.92
	InternVL2-Llama3-76B	34.67	31.95	43.45	42.98	27.70	30.24	54.61	50.15	58.71	63.52	41.45	49.66	54.06	52.02	39.85	35.58	41.56	37.41	34.02	34.82
	GLM-4V-Plus	40.29	25.67	54.05	45.58	29.60	26.76	59.86	59.66	66.48	73.56	58.86	58.44	56.57	50.02	40.88	33.32	37.53	34.64	28.74	33.56
	GPT-4o	28.90	32.65	56.81	46.31	23.73	25.19	71.05	71.78	71.22	75.01	72.01	60.55	74.19	62.81	46.98	35.56	44.57	34.61	27.75	33.65
	Gemini-1.5 Pro	36.24	34.88	52.34	36.21	26.94	26.07	69.85	68.16	75.11	81.17	65.84	58.32	72.46	55.96	42.93	43.01	39.50	36.52	50.46	34.63
	Claude-3.5-Sonnet	31.74	31.63	47.24	38.02	33.77	28.06	67.82	66.82	74.56	81.83	64.97	64.87	72.48	53.77	41.82	40.50	37.77	37.21	42.12	37.42
	o3-mini	36.72	36.23	45.28	34.72	32.62	27.25	70.63	64.52	76.80	57.48	60.77	61.09	64.74	62.99	42.18	42.52	41.97	38.62	44.58	41.31
	Claude-3-7-Sonnet-Thinking	35.85	33.88	53.26	37.78	30.28	27.59	70.38	67.52	74.25	81.12	66.26	65.40	72.81	56.44	38.19	37.74	38.08	37.37	38.19	36.78
	GPT-4.1	29.79	32.13	58.02	39.47	40.03	21.16	69.34	69.12	64.21	64.32	74.83	58.99	76.61	55.15	48.48	38.75	45.73	35.01	49.52	34.42
	Gemini-2.5-flash-preview	34.87	32.65	40.35	33.73	32.30	27.41	71.73	71.04	73.73	64.50	71.55	69.93	71.82	59.42	45.71	42.23	42.14	36.47	55.50	36.14
LLMs (LoRA)	ChatGLM3-6B-Base (LoRA)	35.85	35.43	61.63	55.60	47.80	45.22	66.90	66.24	80.87	79.85	69.60	68.73	74.29	73.96	48.83	45.42	42.62	40.55	46.85	43.57
	LLaMA3-8B-Instruct (LoRA)	39.03	38.86	60.43	55.64	46.87	45.59	74.63	74.46	82.83	81.48	73.06	72.22	71.70	71.44	53.65	51.09	47.25	44.95	50.10	46.81
	InternLM2.5-7B-Chat (LoRA)	35.08	35.06	58.65	50.08	40.96	38.66	65.99	65.26	81.98	80.26	69.65	69.33	67.90	67.73	43.89	42.45	43.89	42.45	46.41	43.36
	LLaMA3.1-8B-Instruct (LoRA)	37.16	37.50	56.86	51.94	40.70	38.38	70.35	69.78	82.65	80.53	67.76	67.23	65.85	65.62	47.16	40.59	41.35	40.46	41.91	39.85
	MiniCPM3-4B-Chat (LoRA)	33.14	32.50	55.89	49.39	39.63	38.61	67.03	65.30	80.44	79.07	68.11	67.22	64.64	63.42	47.13	41.63	42.48	39.77	45.11	40.90
MLLMs (LoRA)	LLaVA1.5-7B-Chat (LoRA)	37.55	35.87	61.05	53.29	41.34	39.24	73.39	72.62	84.39	81.76	76.47	75.73	72.77	72.18	51.83	46.94	43.59	42.16	46.74	43.54
	LLaVA-1.5-13B-Chat (LoRA)	46.60	46.89	67.44	61.53	54.87	53.78	79.72	79.65	85.24	85.95	80.17	80.31	77.65	77.33	58.09	55.06	52.27	51.51	57.33	55.12
	LLaVA-NeXT-7B-Chat (LoRA)	39.06	38.97	60.92	54.35	46.99	42.07	77.99	77.77	85.69	84.39	78.93	78.05	74.64	73.97	52.80	48.38	43.25	41.36	54.16	50.63
	LLaVA-NeXT-13B-Chat (LoRA)	47.41	47.83	66.70	63.42	52.85	51.93	79.59	79.35	85.72	85.87	81.11	81.11	78.83	78.62	59.25	56.63	51.46	50.80	57.19	54.66
	Qwen2-VL-7B-Instruct (LoRA)	38.01	35.68	58.36	57.36	45.40	45.28	78.25	78.44	85.73	86.02	80.60	79.86	78.92	77.78	58.24	54.07	48.94	47.27	56.91	53.04
	Qwen2.5-VL-3B-Instruct (LoRA)	37.06	36.60	60.79	55.33	46.09	44.35	76.13	75.88	84.58	85.82	81.63	81.56	78.16	78.24	60.46	58.07	52.16	52.16	58.49	55.50
	Qwen2.5-VL-7B-Instruct (LoRA)	49.37	48.60	67.17	63.38	54.69	53.42	80.99	80.85	87.60	86.94	82.07	82.02	79.53	79.84	58.44	55.66	49.76	48.47	55.02	51.78
	Llama-3.2-11B-Vision-Instruct (LoRA)	42.56	42.29	61.60	58.82	49.12	49.37	79.11	79.08	83.89	84.70	79.46	79.68	75.17	75.98	62.66	56.77	50.21	47.81	62.10	52.66
MLLMs (MCLF)	LLaVA1.5-7B-Chat (MCLF)	47.74	47.75	64.90	59.40	54.34	50.86	79.84	79.42	84.63	85.24	79.60	79.50	76.72	76.50	63.98	61.52	53.71	53.71	62.79	60.73
	LLaVA-1.5-13B-Chat (MCLF)	48.14	47.45	58.77	62.47	55.11	52.98	79.45	79.78	84.72	84.44	80.76	80.26	77.34	76.99	64.28	61.99	55.24	54.63	64.92	62.41
	LLaVA-NeXT-7B-Chat (MCLF)	46.53	44.73	62.57	60.02	54.22	50.16	78.82	78.29	84.20	83.83	80.60	80.58	76.11	75.97	65.55	61.99	53.79	52.91	64.72	61.41
	LLaVA-NeXT-13B-Chat (MCLF)	48.08	47.61	68.66	61.67	56.15	53.76	79.50	78.50	85.55	85.74	80.90	80.63	78.78	78.30	64.61	62.43	55.53	54.82	62.80	61.31
	Qwen2-VL-7B-Instruct (MCLF)	48.58	48.27	67.47	62.25	54.00	51.75	81.07	80.76	87.22	87.37	82.96	82.54	79.58	79.34	65.38	64.20	56.89	55.37	62.46	62.86
	Qwen2.5-VL-3B-Instruct (MCLF)	45.42	45.85	64.41	60.67	52.74	50.64	80.37	80.26	84.55	85.28	79.87	80.01	75.48	75.48	61.60	59.38	52.21	51.36	60.04	58.73
	Qwen2.5-VL-7B-Instruct (MCLF)	46.60	45.83	68.15	62.47	54.35	52.61	81.78	81.36	86.09	87.34	81.99	81.60	78.62	78.39	65.69	63.62	55.98	54.98	63.64	60.71
	Llama-3.2-11B-Vision-Instruct (MCLF)	48.64	47.73	65.86	61.75	54.44	52.13	80.24	79.55	86.49	86.24	80.86	80.74	77.91	78.08	64.21	60.75	59.78	58.09	66.96	61.89
	Avg. Improvement of MCLF over LoRA	+5.26	+5.26	+2.10	+2.90	+5.50	+4.43	+1.99	+2.66	+0.08	+0.50	+0.89	+0.94	+0.61	+0.64	+6.69	+8.04	+6.44	+6.79	+7.55	+9.14

Table 1: Experimental results of PTMs, LLMs, and MLLMs on metaphor-rich semantic orientation, metaphor-rich pragmatic figurativeness, and partial metaphor reasoning tasks. Cyan means the second best. Purple means the best. Green indicates the average improvement of MCLF over LoRA. *Prec.* and *Rec.* in the table denotes *macro-Prec.* and *macro-Rec.*.

Type	*Model*	*EA*																				
		Happiness			***Love***			***Anger***			***Sorrow***			***Fear***			***Hate***			***Surprise***		
		F1.	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*
LLMs	ChatGLM3-6B-Base	46.32	31.39	88.30	10.24	35.19	5.99	3.01	18.18	1.64	11.76	12.59	11.04	1.83	5.00	1.12	1.54	50.00	0.78	6.27	27.27	3.55
	LLaMA3-8B-Instruct	24.50	44.41	16.91	39.12	37.39	41.01	8.61	4.75	45.90	23.80	18.04	34.97	12.93	10.49	16.85	0.00	0.00	0.00	2.84	35.71	1.48
	InternLM2.5-7B-Chat	50.77	35.71	87.77	20.88	54.19	12.93	19.69	18.94	20.49	30.04	23.10	42.94	6.61	12.50	4.49	2.92	22.22	1.56	8.72	56.14	4.73
	LLaMA3.1-8B-Instruct	46.32	30.98	91.70	10.34	33.92	6.10	5.35	7.69	4.10	10.70	41.67	6.13	0.00	0.00	0.00	0.00	0.00	0.00	1.45	33.33	0.74
	MiniCPM3-4B-Chat	49.43	35.74	80.11	13.47	54.89	7.68	12.94	22.92	9.02	27.33	16.78	73.62	5.31	12.50	3.37	1.23	2.86	0.78	1.17	50.00	0.59
	GPT-3.5-turbo	49.21	40.34	63.09	9.99	57.78	5.47	37.04	33.78	40.98	34.13	28.06	43.56	18.69	16.00	22.47	1.48	14.29	0.78	42.44	35.93	51.85
MLLMs	InternVL2-1B	50.52	36.44	82.34	19.30	47.72	12.09	19.28	13.27	35.25	24.24	22.00	26.99	18.53	14.12	26.97	0.00	0.00	0.00	1.74	54.55	0.89
	InternVL2-2B	49.98	36.71	78.30	23.55	46.44	15.77	23.05	14.90	50.82	27.05	32.20	23.31	15.17	13.11	17.98	1.54	50.00	0.78	7.62	34.52	4.28
	InternVL2-4B	51.61	39.37	74.89	25.49	49.70	17.14	17.35	19.59	15.57	29.16	20.24	52.15	18.40	12.50	34.83	4.08	15.79	2.34	17.95	44.71	11.23
	LLaVA1.5-7B-Chat	14.31	42.19	8.62	37.39	38.00	36.80	8.91	5.30	27.87	10.44	6.44	27.61	0.00	0.00	0.00	11.11	6.72	32.03	0.29	25.00	0.15
	Qwen2-VL-7B-Instruct	50.45	38.06	74.79	12.50	62.86	6.94	17.30	10.86	42.62	25.35	29.75	22.09	15.68	9.28	50.56	0.00	0.00	0.00	3.66	39.39	1.92
	InternVL2-8B	53.37	39.73	81.28	27.66	57.67	18.19	26.96	21.83	35.25	36.29	26.48	57.67	19.25	18.37	20.22	6.02	80.00	3.12	22.32	50.52	14.33
	Llama-3.2-11B-Vision-Instruct	46.56	36.47	64.36	29.77	37.68	24.61	15.69	13.04	19.67	26.21	18.53	44.79	16.83	15.04	19.10	11.11	13.64	9.38	1.45	45.45	0.74
	InternVL2-26B	52.74	41.05	73.72	29.57	53.74	20.40	30.83	31.36	30.33	32.99	28.44	39.26	24.71	15.59	59.55	8.11	30.00	4.69	34.57	54.09	25.41
	InternVL2-40B	53.24	40.95	76.06	29.99	60.13	19.98	27.13	25.74	28.69	35.84	24.82	64.42	23.17	17.65	33.71	18.58	30.91	13.28	27.52	55.36	18.32
	Qwen2-VL-72B-Instruct	50.84	38.20	75.96	32.24	51.99	23.37	27.01	24.34	30.33	34.38	23.53	63.80	15.62	25.64	11.24	0.00	0.00	0.00	17.16	50.36	10.34
	InternVL2-Llama3-76B	51.70	41.35	68.94	26.58	50.15	18.09	14.81	12.57	18.03	34.38	23.97	60.74	15.66	11.46	24.72	11.19	53.33	6.25	34.93	49.86	26.88
	GLM-4V-Plus	48.02	36.97	68.51	35.27	48.35	27.76	0.00	0.00	0.00	29.53	19.10	65.03	9.80	38.46	5.62	1.55	100.00	0.78	18.33	39.13	11.96
	GPT-4o	48.54	39.53	62.87	40.39	47.91	34.91	19.07	14.29	28.69	38.17	27.23	63.80	23.26	19.84	28.09	0.00	0.00	0.00	17.12	53.49	10.19
	Gemini-1.5 Pro	52.39	39.13	79.26	25.45	55.48	16.51	25.23	29.35	22.13	41.54	29.66	69.33	29.52	25.62	34.83	2.94	25.00	1.56	29.02	49.47	20.53
	Claude-3.5-Sonnet	51.06	40.01	70.53	29.03	46.31	21.14	15.30	22.95	11.48	34.73	25.21	55.83	26.54	22.95	31.46	11.90	25.00	7.81	29.29	39.75	23.19
	o3-mini	52.82	40.86	74.68	32.28	57.07	22.50	21.00	26.92	17.21	33.70	26.39	46.63	30.97	20.21	66.29	4.38	33.33	2.34	32.83	52.26	23.93
	Claude-3-7-Sonnet-Thinking	51.82	40.24	72.77	27.21	59.15	17.67	14.29	26.09	9.84	37.83	26.47	66.26	24.71	18.82	35.96	11.84	37.50	7.03	33.54	42.69	27.62
	GPT-4.1	45.44	40.40	51.91	46.77	44.38	49.42	9.36	16.33	6.56	34.90	22.93	73.01	27.80	23.13	34.83	0.00	0.00	0.00	15.94	61.39	9.16
	Gemini-2.5-flash-preview	53.70	40.15	81.06	23.05	65.52	13.99	10.67	28.57	6.56	38.30	24.91	82.82	20.74	30.43	15.73	0.00	0.00	0.00	37.32	54.55	28.36
LLMs (LoRA)	ChatGLM3-6B-Base (LoRA)	45.31	45.41	45.21	43.37	43.22	43.53	33.93	37.25	31.15	29.05	26.67	31.90	23.16	21.78	24.72	27.12	29.63	25.00	46.77	47.01	46.53
	LLaMA3-8B-Instruct (LoRA)	49.79	50.00	49.57	46.50	46.95	46.06	33.64	37.76	30.33	35.10	32.14	38.65	26.74	25.51	28.09	28.81	30.43	27.34	51.20	50.43	51.99
	InternLM2.5-7B-Chat (LoRA)	47.24	46.32	48.19	43.80	43.76	43.85	31.54	31.93	31.15	33.80	31.25	36.81	20.12	21.25	19.10	24.90	25.62	24.22	43.71	45.45	42.10
	LLaMA3.1-8B-Instruct (LoRA)	45.59	45.86	45.32	44.97	45.80	44.16	35.10	34.96	35.25	35.16	33.15	37.42	23.60	26.39	21.35	30.26	28.67	32.03	46.12	45.30	46.97
	MiniCPM3-4B-Chat (LoRA)	45.15	42.98	47.55	39.98	39.69	40.27	31.49	32.74	30.33	33.14	31.49	34.97	20.11	20.00	20.22	17.62	20.20	15.62	41.46	44.85	38.55
MLLMs (LoRA)	LLaVA1.5-7B-Chat (LoRA)	43.60	43.70	43.51	43.69	41.72	45.85	35.98	36.75	35.25	31.16	28.95	33.74	24.16	30.00	20.22	28.57	30.91	26.56	48.25	50.82	45.94
	LLaVA-1.5-13B-Chat (LoRA)	57.36	57.70	57.02	54.16	54.59	53.73	43.88	45.22	42.62	45.14	42.25	48.47	33.67	30.84	37.08	35.34	39.42	32.03	56.73	56.15	57.31
	LLaVA-NeXT-7B-Chat (LoRA)	48.96	50.34	47.66	46.43	45.98	46.90	41.35	42.61	40.16	31.86	30.68	33.13	25.14	25.58	24.72	28.26	26.35	30.47	50.79	51.85	49.78
	LLaVA-NeXT-13B-Chat (LoRA)	57.36	57.70	57.02	54.16	54.59	53.73	43.88	45.22	42.62	45.14	42.25	48.47	33.67	30.84	37.08	35.34	39.42	32.03	56.73	56.15	57.31
	Qwen2-VL-7B-Instruct (LoRA)	46.86	43.22	51.17	42.27	41.07	43.53	32.41	37.23	28.69	36.41	32.68	41.10	27.27	32.31	23.60	26.40	27.05	25.78	42.63	52.48	35.89
	Qwen2.5-VL-3B-Instruct (LoRA)	45.76	46.20	45.32	44.63	42.49	47.00	34.60	35.65	33.61	36.41	33.51	39.88	21.66	25.00	19.10	24.35	23.08	25.78	49.16	53.47	45.49
	Qwen2.5-VL-7B-Instruct (LoRA)	56.30	56.97	55.64	53.76	54.22	53.31	43.48	52.94	36.89	48.86	45.50	52.76	41.67	44.30	39.33	39.29	36.18	42.97	56.88	55.48	58.35
	Llama-3.2-11B-Vision-Instruct (LoRA)	52.37	53.75	51.06	48.79	48.79	48.79	40.18	44.12	36.89	36.60	31.56	43.56	33.73	36.36	31.46	31.50	31.75	31.25	52.29	51.58	53.03
MLLMs (MCLF)	LLaVA1.5-7B-Chat (MCLF)	57.90	59.16	56.70	55.10	54.78	55.42	41.67	42.37	40.98	40.87	41.25	40.49	41.18	36.52	47.19	38.26	43.14	34.38	58.01	56.98	59.08
	LLaVA-1.5-13B-Chat (MCLF)	57.84	58.79	56.91	54.78	55.31	54.26	45.41	48.60	42.62	42.20	39.89	44.79	41.18	43.21	39.33	35.86	36.59	35.16	56.74	54.57	59.08
	LLaVA-NeXT-7B-Chat (MCLF)	58.55	59.58	57.55	55.02	56.26	53.84	45.45	51.02	40.98	38.18	37.72	38.65	31.76	33.33	30.34	35.16	35.16	35.16	54.52	52.61	56.57
	LLaVA-NeXT-13B-Chat (MCLF)	59.11	59.40	58.83	55.84	55.64	56.05	42.11	45.28	39.34	45.45	44.91	46.01	33.33	31.07	35.96	40.33	42.61	38.28	58.23	57.68	58.79
	Qwen2-VL-7B-Instruct (MCLF)	57.13	58.00	56.28	55.12	56.81	53.52	40.51	41.74	39.34	47.46	43.98	51.53	39.26	43.24	35.96	40.64	41.46	39.84	57.94	54.81	61.45
	Qwen2.5-VL-3B-Instruct (MCLF)	55.53	57.83	53.40	53.34	52.16	54.57	44.64	46.85	42.62	39.05	37.71	40.49	35.90	33.02	39.33	35.57	36.00	35.16	54.86	54.35	55.39
	Qwen2.5-VL-7B-Instruct (MCLF)	57.88	58.32	57.45	55.68	55.74	55.63	33.80	39.56	29.51	43.80	41.30	46.63	35.23	35.63	34.83	39.04	39.84	38.28	57.10	55.77	58.49
	Llama-3.2-11B-Vision-Instruct (MCLF)	59.08	59.33	58.83	55.46	55.08	55.84	44.44	48.54	40.98	45.71	42.78	49.08	36.99	38.10	35.96	38.52	40.52	36.72	56.43	56.14	56.72

Table 2: Experimental results of LLMs, and MLLMs on EA tasks.

Type	Model	OD									IR														
		No_Offensive			Slightly			Moderately			Interactive			Expressive			Entertaining			Offensive			Other		
		F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.	F1.	Prec.	Rec.
LLMs	ChatGLM3-6B-Base	85.18	81.92	88.70	16.27	18.62	14.44	5.56	13.16	3.52	21.06	13.26	51.18	26.47	26.47	26.47	0.88	33.33	0.44	13.71	8.61	33.59	2.02	10.53	1.12
	LLaMA3-8B-Instruct	82.24	84.01	80.55	23.03	20.13	26.89	15.32	19.35	12.68	3.02	7.55	1.89	43.16	28.04	93.65	12.18	62.81	6.74	0.00	0.00	0.00	0.00	0.00	0.00
	InternLM2.5-7B-Chat	89.07	83.56	95.36	15.44	44.68	9.33	20.00	19.59	20.42	16.45	13.72	20.52	22.33	35.19	16.35	37.70	43.78	33.10	2.84	20.00	1.53	25.98	18.91	41.45
	LLaMA3.1-8B-Instruct	83.66	81.41	86.04	11.45	16.88	8.67	6.08	5.00	7.75	23.29	13.49	85.14	15.87	25.45	11.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	MiniCPM3-4B-Chat	87.61	85.46	89.87	21.18	27.80	17.11	27.36	24.06	31.69	12.01	15.83	9.67	40.72	39.93	41.53	15.00	54.44	8.70	22.80	35.48	16.79	28.97	19.11	59.85
	GPT-3.5-turbo	83.93	85.42	82.49	17.86	20.58	15.78	18.14	12.95	30.28	18.14	16.90	19.58	19.20	43.03	12.35	25.94	50.26	17.48	16.95	32.61	11.45	28.17	18.08	63.75
MLLMs	InternVL2-1B	88.63	81.92	96.53	13.64	31.97	8.67	1.18	3.57	0.70	0.45	4.17	0.24	0.00	0.00	0.00	22.22	39.06	15.53	7.50	4.06	48.09	25.47	19.27	37.55
	InternVL2-2B	44.28	92.21	29.14	26.96	16.23	79.56	17.04	23.46	13.38	1.37	23.08	0.71	36.90	31.53	44.47	42.41	43.73	41.17	24.34	18.62	35.11	22.43	22.18	22.68
	InternVL2-4B	89.46	86.48	92.66	32.43	36.26	29.33	14.51	27.45	9.86	5.79	10.43	4.01	44.59	31.13	78.59	36.18	53.18	27.42	21.80	21.48	22.14	4.12	26.67	2.23
	LLaVA1.5-7B-Chat	61.63	86.27	47.94	26.66	18.49	47.78	8.04	5.09	19.01	14.74	15.98	13.68	34.97	27.74	47.29	39.36	41.60	37.36	6.38	4.90	9.16	0.00	0.00	0.00
	Qwen2-VL-7B-Instruct	85.56	84.70	86.44	10.51	42.19	6.00	17.77	11.53	38.73	10.15	14.60	7.78	9.26	31.94	5.41	55.38	39.80	91.04	5.71	44.44	3.05	10.45	30.09	6.32
	InternVL2-8B	89.37	80.84	99.92	0.44	25.00	0.22	1.38	33.33	0.70	3.26	11.94	1.89	41.02	34.34	50.94	53.49	50.55	56.79	25.14	19.82	34.35	20.08	31.73	14.68
	Llama-3.2-11B-Vision-Instruct	86.33	89.27	83.58	38.88	33.44	46.44	17.23	18.40	16.20	12.22	21.82	8.49	41.20	30.97	61.53	46.36	46.95	45.79	13.89	17.65	11.45	4.21	37.50	2.23
	InternVL2-26B	89.67	81.60	99.52	6.61	47.06	3.56	8.97	50.00	4.93	2.21	17.86	1.18	30.14	38.36	24.82	46.37	51.01	42.50	32.84	26.96	41.98	32.64	22.83	57.25
	InternVL2-40B	89.97	84.56	96.13	17.74	32.35	12.22	23.11	31.33	18.31	1.37	23.08	0.71	40.48	36.29	45.76	51.71	53.79	49.78	11.54	36.00	6.87	30.65	24.32	41.45
	Qwen2-VL-72B-Instruct	90.10	83.57	97.74	13.89	31.75	8.89	21.28	43.48	14.08	5.42	23.21	3.07	36.67	31.43	44.00	61.33	49.72	80.02	2.82	18.18	1.53	0.00	0.00	0.00
	InternVL2-Llama3-76B	79.65	85.90	74.25	23.93	18.60	33.56	23.26	25.86	21.13	5.78	10.37	4.01	42.53	31.53	65.29	42.14	52.23	35.31	29.61	23.35	40.46	9.50	21.02	6.13
	GLM-4V-Plus	91.25	87.17	95.72	36.68	46.13	30.44	15.46	28.85	10.56	9.86	17.68	6.84	26.99	38.77	20.71	47.58	40.76	57.14	21.26	28.95	16.79	26.09	21.86	32.34
	GPT-4o	91.33	87.59	95.40	36.50	43.29	31.56	18.38	39.53	11.97	0.90	11.11	0.47	45.01	30.62	84.94	48.00	64.42	38.24	3.87	12.50	2.29	0.00	0.00	0.00
	Gemini-1.5 Pro	89.83	82.37	98.79	14.02	41.30	8.44	2.70	33.33	1.41	3.16	36.84	1.65	42.79	31.35	67.41	58.44	56.50	60.51	1.42	10.00	0.76	0.00	0.00	0.00
	Claude-3.5-Sonnet	90.19	83.82	97.62	18.09	34.81	12.22	7.14	23.08	4.23	18.79	19.74	17.92	43.72	33.23	63.88	55.93	59.01	53.15	7.79	26.09	4.58	1.45	30.77	0.74
	o3-mini	89.60	81.56	99.39	4.95	34.29	2.67	3.82	20.00	2.11	18.15	24.60	14.39	45.08	34.16	66.24	45.37	63.72	35.23	6.67	26.32	3.82	17.38	17.48	17.29
	Claude-3-7-Sonnet-Thinking	89.84	83.09	97.78	17.32	36.69	11.33	7.64	40.00	4.23	19.24	20.77	17.92	43.15	32.96	62.47	57.74	59.11	56.43	1.42	10.00	0.76	0.73	28.57	0.37
	GPT-4.1	90.69	84.22	98.22	24.23	44.38	16.67	6.54	45.45	3.52	0.94	66.67	0.47	43.95	28.23	99.18	10.14	80.26	5.41	1.48	25.00	0.76	0.00	0.00	0.00
	Gemini-2.5-flash-preview	89.37	81.05	99.60	1.70	20.00	0.89	1.36	20.00	0.70	0.93	28.57	0.47	46.47	32.40	82.12	38.86	68.30	27.15	0.00	0.00	0.00	29.58	32.24	27.32
LLMs (LoRA)	ChatGLM3-6B-Base (LoRA)	92.35	90.48	94.31	49.23	52.38	46.44	32.17	42.05	26.06	40.46	44.60	37.03	48.39	49.03	47.76	66.14	61.85	71.07	32.46	38.14	28.24	43.63	45.38	42.01
	LLaMA3-8B-Instruct (LoRA)	92.47	91.27	93.70	51.06	51.47	50.67	28.44	38.55	22.54	43.11	45.99	40.57	45.36	48.53	42.59	68.07	62.64	74.53	29.60	31.09	28.24	43.97	46.12	42.01
	InternLM2.5-7B-Chat (LoRA)	92.03	88.96	95.32	46.12	51.80	41.56	19.39	35.19	13.38	30.29	38.41	25.00	41.80	43.36	40.35	62.99	56.62	70.98	20.00	23.23	17.56	41.21	43.18	39.41
	LLaMA3.1-8B-Instruct (LoRA)	92.09	90.49	93.74	48.65	49.32	48.00	19.32	30.77	14.08	34.65	41.10	29.95	39.50	40.07	38.94	61.20	56.15	67.26	23.48	27.27	20.61	36.91	38.89	35.13
	MiniCPM3-4B-Chat (LoRA)	91.68	88.80	94.75	43.81	49.44	39.33	19.05	29.41	14.08	24.12	34.20	18.63	39.31	39.81	38.82	61.47	56.66	67.17	29.32	28.89	29.77	38.63	38.59	38.66
MLLMs (LoRA)	LLaVA1.5-7B-Chat (LoRA)	92.57	90.67	94.55	50.23	51.40	49.11	23.23	41.07	16.20	30.39	35.78	26.42	43.07	42.63	43.53	62.79	58.28	68.06	25.00	28.71	22.14	38.49	41.28	36.06
	LLaVA-1.5-13B-Chat (LoRA)	93.84	92.68	95.04	57.21	57.99	56.44	40.34	51.65	33.10	53.03	53.48	52.59	53.84	55.42	52.35	72.35	69.68	75.24	39.02	41.74	36.64	53.03	54.05	52.04
	LLaVA-NeXT-7B-Chat (LoRA)	92.71	91.34	94.11	51.22	51.11	51.33	24.51	40.32	17.61	38.51	44.07	34.20	44.54	46.82	42.47	64.99	62.49	67.70	30.38	33.96	27.48	42.55	47.59	38.48
	LLaVA-NeXT-13B-Chat (LoRA)	93.60	93.34	93.87	58.23	56.75	59.78	42.28	50.00	36.62	47.69	49.25	46.23	51.34	51.99	50.71	70.84	68.93	72.85	41.43	43.33	39.69	50.47	50.75	50.19
	Qwen2-VL-7B-Instruct (LoRA)	92.96	92.98	92.94	52.10	50.52	53.78	28.12	31.58	25.35	40.58	45.97	36.32	43.39	50.15	38.24	67.68	61.68	74.98	28.47	25.61	32.06	44.18	43.58	44.80
	Qwen2.5-VL-3B-Instruct (LoRA)	92.69	91.54	93.87	49.72	50.57	48.89	29.46	40.24	23.24	40.58	46.36	36.08	45.58	46.25	44.94	66.41	63.39	69.74	30.00	33.03	27.48	42.43	41.42	43.49
	Qwen2.5-VL-7B-Instruct (LoRA)	93.99	93.95	94.03	58.05	55.47	60.89	42.02	52.08	35.21	51.61	52.30	50.94	53.49	54.81	52.24	73.04	71.51	74.62	41.67	45.87	38.17	50.00	48.93	51.12
	Llama-3.2-11B-Vision-Instruct (LoRA)	93.34	93.31	93.38	55.37	52.60	58.44	30.17	38.89	24.65	47.42	49.49	45.52	46.58	52.27	42.00	66.91	62.20	72.40	37.32	34.64	40.46	46.73	46.99	46.47
MLLMs (MCLF)	LLaVA1.5-7B-Chat (MCLF)	92.01	92.74	91.28	56.03	54.78	57.33	36.36	47.19	29.58	47.41	50.95	44.34	53.12	52.28	54.00	70.48	67.87	73.29	40.00	49.44	33.59	50.09	51.16	49.07
	LLaVA-1.5-13B-Chat (MCLF)	93.37	92.68	94.07	56.02	55.17	56.89	30.90	39.56	25.35	48.69	49.04	48.35	52.86	54.80	51.06	72.63	69.80	75.69	42.61	49.49	37.40	52.42	52.42	52.42
	LLaVA-NeXT-7B-Chat (MCLF)	93.29	92.85	93.74	54.73	54.13	55.33	35.20	40.74	30.99	50.81	53.52	48.35	52.59	53.33	51.88	69.82	67.81	71.96	35.41	47.44	28.24	49.68	49.01	50.37
	LLaVA-NeXT-13B-Chat (MCLF)	93.76	92.67	94.87	58.44	58.44	58.44	40.18	54.88	31.69	50.95	54.74	47.64	55.24	54.73	55.76	72.83	69.72	76.22	44.81	49.09	41.22	50.10	52.44	47.96
	Qwen2-VL-7B-Instruct (MCLF)	93.90	93.34	94.47	60.19	58.49	62.00	37.89	50.59	30.28	52.08	54.06	50.24	51.37	53.22	49.65	72.44	69.17	76.04	37.00	43.75	32.06	50.28	49.82	50.74
	Qwen2.5-VL-3B-Instruct (MCLF)	93.74	93.15	94.35	55.92	55.19	56.67	36.67	44.90	30.99	49.05	49.52	48.58	50.18	50.97	49.41	69.27	67.80	70.81	41.23	48.45	35.88	47.71	46.94	48.51
	Qwen2.5-VL-7B-Instruct (MCLF)	93.79	93.75	93.83	59.81	56.14	64.00	38.36	54.55	29.58	50.18	51.10	49.29	52.42	53.85	51.06	72.63	70.73	74.62	38.96	45.00	34.35	52.36	51.06	53.72
	Llama-3.2-11B-Vision-Instruct (MCLF)	93.84	93.93	93.74	59.13	55.45	63.33	35.56	48.19	28.17	47.74	51.08	44.81	55.16	56.00	54.35	72.77	69.03	76.93	39.15	44.23	35.11	50.62	51.85	49.44

Table 3: Experimental results of LLMs, and MLLMs on OD and IR tasks.

Type	*Model*	*HumorD*						*SD*						*HyperD*						*MD*					
		True			***False***			***True***			***False***			***True***			***False***			***True***			***False***		
		F1.	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*
LLMs	ChatGLM3-6B-Base	71.31	61.93	84.03	33.98	52.15	25.20	32.14	21.16	66.78	57.14	84.98	43.03	66.47	60.52	73.72	34.17	42.34	28.64	75.31	63.75	92.00	16.80	43.22	10.42
	LLaMA3-8B-Instruct	70.58	62.87	80.45	39.10	52.41	31.18	34.11	23.39	62.94	65.48	86.15	52.80	61.57	61.42	61.72	42.63	42.79	42.48	72.67	64.54	83.13	28.96	43.03	21.82
	InternLM2.5-7B-Chat	61.12	66.11	56.83	52.48	48.05	57.81	34.63	27.46	46.85	77.96	85.48	71.66	57.51	63.24	52.73	48.54	43.74	54.53	68.44	66.06	71.00	40.11	43.06	37.54
	LLaMA3.1-8B-Instruct	73.23	95.04	59.56	11.50	47.67	6.54	31.05	18.58	94.41	9.92	80.49	5.28	73.04	59.77	93.89	10.81	40.74	6.23	75.71	63.19	94.43	10.11	37.93	5.83
	MiniCPM3-4B-Chat	40.64	66.33	29.30	55.88	43.39	78.47	33.37	40.60	28.32	87.48	84.65	90.51	33.38	71.63	21.76	57.51	42.90	87.22	65.52	66.97	64.14	44.25	42.75	45.85
	GPT-3.5-turbo	53.91	74.16	42.35	59.99	48.50	78.63	29.02	22.65	40.38	75.15	83.37	68.41	58.71	56.57	61.01	32.39	34.52	30.50	47.54	45.75	49.47	67.34	69.01	65.74
MLLMs	InternVL2-1B	0.00	0.00	0.00	57.41	40.26	100.00	0.00	0.00	0.00	89.71	81.36	99.96	8.12	36.24	4.57	54.11	39.00	88.36	17.03	49.87	10.27	49.01	34.89	82.33
	InternVL2-2B	60.55	57.14	64.39	31.26	34.90	28.32	18.24	57.41	10.84	89.82	82.78	98.16	69.87	60.81	82.10	31.30	47.41	23.37	57.68	66.58	50.88	46.82	40.09	56.27
	InternVL2-4B	71.36	61.18	85.61	27.59	47.62	19.42	32.00	19.16	96.85	12.03	89.94	6.45	60.58	71.04	52.81	58.04	50.17	68.82	77.50	63.35	99.79	2.26	76.47	1.15
	LLaVA1.5-7B-Chat	72.99	59.22	95.10	9.23	42.21	5.18	30.81	19.31	76.22	40.85	83.25	27.06	60.30	60.14	60.47	40.70	40.86	40.53	74.71	62.86	92.05	11.44	33.62	6.89
	Qwen2-VL-7B-Instruct	48.88	71.55	37.11	58.31	46.33	78.63	33.30	34.13	32.52	85.17	84.71	85.63	32.81	68.28	21.59	56.47	42.25	85.11	56.21	66.74	48.56	47.49	39.94	58.57
	InternVL2-8B	62.72	66.26	59.54	51.17	47.82	55.02	53.80	42.81	72.38	84.55	92.49	77.86	59.93	77.12	49.01	62.46	51.67	78.95	75.71	65.13	90.40	25.66	51.05	17.14
	Llama-3.2-11B-Vision-Instruct	52.12	69.16	41.82	55.93	45.59	72.33	38.16	28.42	58.04	75.55	87.38	66.53	44.92	69.88	33.09	57.45	45.02	79.35	1.22	53.85	0.62	77.36	63.20	99.69
	InternVL2-26B	71.69	73.39	70.07	60.27	58.38	62.30	55.17	44.24	73.25	85.26	92.79	78.86	76.60	71.37	82.65	58.71	67.43	51.99	78.57	65.65	97.83	21.31	76.92	12.37
	InternVL2-40B	65.56	67.49	63.74	52.29	50.30	54.45	53.70	38.26	90.03	78.97	96.69	66.73	74.07	76.85	71.48	65.50	62.49	68.82	78.42	65.37	97.99	19.43	76.36	11.13
	Qwen2-VL-72B-Instruct	71.81	74.35	69.44	62.30	59.58	65.28	26.96	63.23	17.13	90.19	83.73	97.72	48.81	77.30	35.66	60.32	46.92	84.45	78.43	66.63	95.30	28.85	69.36	18.21
	InternVL2-Llama3-76B	74.31	59.57	98.75	1.11	23.33	0.57	38.82	25.24	84.09	58.60	92.18	42.95	74.21	59.23	99.34	1.88	50.00	0.96	74.29	64.21	88.13	23.35	43.90	15.90
	GLM-4V-Plus	68.02	66.76	69.33	51.44	52.96	50.00	51.84	41.13	70.10	83.78	91.84	77.02	53.11	71.15	42.37	57.31	46.56	74.51	77.39	63.14	99.95	0.18	50.00	0.09
	GPT-4o	73.39	80.30	67.57	68.17	61.80	76.00	56.88	51.19	63.99	88.56	91.25	86.03	77.67	65.59	95.20	38.93	78.43	25.89	80.68	70.09	95.05	43.96	78.28	30.57
	Gemini-1.5 Pro	61.39	85.77	47.80	67.03	53.94	88.52	64.58	56.22	75.87	90.08	93.99	86.47	75.75	64.44	91.88	36.19	67.25	24.76	78.88	66.10	97.78	23.97	78.82	14.13
	Claude-3.5-Sonnet	61.37	82.24	48.95	65.49	53.39	84.69	64.42	54.55	78.67	89.52	94.57	84.99	72.01	71.57	72.46	57.82	58.37	57.28	15.91	80.00	8.83	78.35	64.96	98.71
	o3-mini	49.13	91.45	33.59	65.46	49.81	95.45	26.84	69.85	16.61	90.46	83.74	98.36	61.64	71.54	54.14	57.64	50.00	68.04	75.92	71.62	80.75	50.77	57.85	45.23
	Claude-3-7-Sonnet-Thinking	58.40	87.93	43.72	66.94	52.84	91.31	63.73	54.31	77.10	89.44	94.20	85.15	74.11	71.15	77.32	57.16	61.37	53.48	79.02	66.35	97.68	25.50	79.26	15.19
	GPT-4.1	66.07	82.58	55.07	67.01	56.11	83.17	41.91	41.70	42.13	86.61	86.72	86.51	77.69	64.53	97.60	32.90	85.14	20.39	78.96	65.64	99.07	19.89	87.59	11.22
	Gemini-2.5-flash-preview	66.85	86.39	54.52	69.10	57.07	87.56	43.64	61.20	33.92	90.46	86.27	95.08	78.16	74.13	82.66	62.54	68.98	57.20	79.53	68.07	95.61	35.54	75.57	23.23
LLMs (LoRA)	ChatGLM3-6B-Base (LoRA)	74.00	71.53	76.65	58.87	62.28	55.82	67.86	69.34	66.43	92.83	92.39	93.27	76.43	73.77	79.28	61.58	65.42	58.17	81.21	80.49	81.94	67.03	68.09	65.99
	LLaMA3-8B-Instruct (LoRA)	79.42	78.84	80.01	69.65	70.42	68.90	70.74	72.69	68.88	93.51	92.96	94.08	78.91	76.56	81.41	66.13	69.55	63.03	79.32	78.68	79.98	63.80	64.73	62.90
	InternLM2.5-7B-Chat (LoRA)	73.43	70.69	76.38	57.49	61.28	54.15	68.96	71.48	66.61	93.19	92.47	93.92	75.96	74.85	77.10	62.97	64.44	61.57	76.63	75.87	77.40	58.88	59.93	57.86
	LLaMA3.1-8B-Instruct (LoRA)	76.43	74.49	78.47	63.54	66.21	61.08	69.64	72.76	66.78	93.40	92.53	94.28	74.79	72.92	76.77	60.04	62.60	57.69	75.13	74.43	75.85	56.31	57.26	55.39
	MiniCPM3-4B-Chat (LoRA)	74.96	70.00	80.67	56.12	64.07	49.92	66.79	68.83	64.86	92.66	92.06	93.27	75.40	72.57	78.46	59.58	63.66	55.99	75.45	72.16	79.05	52.04	57.13	47.79
MLLMs (LoRA)	LLaVA1.5-7B-Chat (LoRA)	78.85	76.56	81.28	66.94	70.23	63.96	71.99	75.82	68.53	93.96	92.95	95.00	81.40	79.48	83.42	70.64	73.45	68.04	80.26	78.94	81.63	64.60	66.60	62.72
	LLaVA-1.5-13B-Chat (LoRA)	83.47	83.24	83.70	75.90	76.21	75.60	76.62	75.64	77.62	94.56	94.84	94.28	83.98	84.37	83.59	76.50	75.98	77.02	83.61	82.98	84.26	71.35	72.32	70.41
	LLaVA-NeXT-7B-Chat (LoRA)	82.14	81.47	82.82	73.61	74.51	72.73	75.42	77.15	73.78	94.52	94.05	95.00	83.30	81.17	85.55	73.49	76.69	70.55	81.46	80.30	82.66	66.97	68.74	65.28
	LLaVA-NeXT-13B-Chat (LoRA)	83.43	82.73	84.14	75.49	76.45	74.56	76.90	76.70	77.10	94.69	94.75	94.64	84.79	84.79	84.79	77.43	77.43	77.43	84.45	84.01	84.88	72.99	73.65	72.35
	Qwen2-VL-7B-Instruct (LoRA)	82.10	82.76	81.44	74.58	73.73	75.44	77.04	76.64	77.45	94.71	94.82	94.60	84.56	82.77	86.42	75.78	78.44	73.30	84.64	82.65	86.74	71.86	75.19	68.82
	Qwen2.5-VL-3B-Instruct (LoRA)	80.68	79.87	81.50	71.31	72.39	70.26	76.00	74.29	77.80	94.34	94.86	93.84	85.23	85.02	85.44	77.95	78.24	77.67	83.87	84.04	83.69	72.54	72.28	72.79
	Qwen2.5-VL-7B-Instruct (LoRA)	84.52	84.12	84.91	77.32	77.85	76.79	79.22	80.14	78.32	95.31	95.06	95.56	85.57	85.43	85.71	78.51	78.70	78.32	84.82	85.48	84.16	74.54	73.58	75.53
	Llama-3.2-11B-Vision-Instruct (LoRA)	82.95	82.86	83.04	75.24	75.36	75.12	75.30	72.95	77.80	94.11	94.84	93.39	83.34	84.04	82.66	75.78	74.88	76.70	81.11	83.34	79.00	69.85	66.99	72.97
MLLMs (MCLF)	LLaVA1.5-7B-Chat (MCLF)	83.69	82.50	84.91	75.52	77.19	73.92	75.54	74.70	76.40	94.32	94.57	94.08	83.62	83.32	83.91	75.48	75.88	75.08	82.91	82.45	83.38	70.29	70.99	69.61
	LLaVA-1.5-13B-Chat (MCLF)	83.61	82.67	84.58	75.59	76.90	74.32	74.87	75.27	74.48	94.28	94.17	94.40	84.64	83.39	85.93	76.32	78.14	74.60	83.40	82.70	84.11	70.91	71.97	69.88
	LLaVA-NeXT-7B-Chat (MCLF)	82.91	81.46	84.42	74.12	76.18	72.17	73.94	74.47	73.43	94.08	93.93	94.24	84.39	84.32	84.46	76.79	76.89	76.70	82.45	82.13	82.77	69.63	70.10	69.17
	LLaVA-NeXT-13B-Chat (MCLF)	83.53	81.05	86.18	74.22	77.96	70.81	76.66	76.39	76.92	94.63	94.71	94.56	84.70	83.98	85.44	76.80	77.82	75.81	84.46	83.54	85.40	72.58	74.01	71.20
	Qwen2-VL-7B-Instruct (MCLF)	84.63	83.76	85.52	77.17	78.37	76.00	79.34	79.13	79.55	79.34	79.13	79.55	86.35	85.31	87.40	79.11	80.60	77.67	84.99	84.53	85.45	73.92	74.62	73.23
	Qwen2.5-VL-3B-Instruct (MCLF)	84.01	83.67	84.36	76.61	77.08	76.16	75.52	74.49	76.57	94.30	94.60	94.00	83.73	84.14	83.32	76.14	75.60	76.70	81.91	81.93	81.89	69.05	69.02	69.08
	Qwen2.5-VL-7B-Instruct (MCLF)	85.23	84.08	86.40	77.87	79.49	76.32	78.46	76.76	80.24	94.93	95.43	94.44	85.58	84.60	86.59	77.97	79.38	76.62	84.30	83.83	84.78	72.70	73.42	72.00
	Llama-3.2-11B-Vision-Instruct (MCLF)	84.05	82.28	85.90	75.62	78.19	73.21	77.79	78.13	77.45	94.94	94.85	95.04	84.63	84.31	84.95	76.97	77.41	76.54	83.65	84.02	83.28	72.34	71.80	72.88

Table 4: Experimental results of LLMs, and MLLMs on metaphor-rich pragmatic figurativeness and MD tasks.

Type	Model	SDD									TDD									MC								
		Text			Image			Complementary			Text			Image			Complementary			Text			Image			Complementary		
		F1.	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*	*F1.*	*Prec.*	*Rec.*
LLMs	ChatGLM3-6B-Base	31.18	18.68	94.10	9.84	53.27	5.42	2.46	18.42	1.32	48.91	33.07	93.83	7.01	19.63	4.27	2.82	31.58	1.47	26.72	15.63	91.89	8.71	20.45	5.53	7.05	65.15	3.73
	LLaMA3-8B-Instruct	27.21	17.20	65.17	3.15	60.71	1.62	29.70	28.88	30.57	34.38	29.37	41.46	5.61	20.51	3.25	44.78	41.22	49.02	23.97	15.10	58.11	5.64	34.09	3.07	47.98	60.66	39.69
	InternLM2.5-7B-Chat	30.84	18.37	96.07	1.69	64.29	0.86	3.72	17.74	2.08	49.00	32.81	96.68	1.98	35.71	1.02	5.25	37.10	2.83	26.39	15.25	97.97	0.81	66.67	0.41	2.86	51.52	1.47
	LLaMA3.1-8B-Instruct	30.99	18.57	93.54	1.87	58.82	0.95	10.94	28.12	6.79	48.16	32.89	89.87	2.66	20.59	1.42	14.53	40.68	8.85	27.04	15.75	95.61	1.18	15.00	0.61	12.71	66.94	7.02
	MiniCPM3-4B-Chat	31.03	18.40	98.88	0.00	0.00	0.00	3.61	41.67	1.89	49.19	32.72	99.05	0.00	0.00	0.00	3.10	54.17	1.60	26.43	15.25	98.99	0.81	40.00	0.41	1.20	0.61	58.33
	GPT-3.5-turbo	30.59	18.14	97.47	0.00	0.00	0.00	3.96	44.00	2.08	49.24	32.73	99.37	0.00	0.00	0.00	3.36	73.68	1.72	26.52	15.29	100.00	0.00	0.00	0.00	0.35	100.00	0.17
MLLMs	InternVL2-1B	1.10	25.00	0.56	0.57	37.50	0.29	42.74	27.26	98.87	0.00	0.00	0.00	40.35	25.31	99.39	0.24	20.00	0.12	1.31	22.22	0.68	40.36	25.30	99.80	0.69	100.00	0.35
	InternVL2-2B	1.08	14.29	0.56	66.57	55.53	83.08	26.82	33.71	22.26	0.00	0.00	0.00	40.51	25.40	100.00	0.25	100.00	0.12	0.00	0.00	0.00	29.22	29.34	29.10	66.62	59.94	74.96
	InternVL2-4B	8.17	28.33	4.78	70.38	54.90	98.00	0.00	0.00	0.00	9.90	25.00	6.17	0.00	0.00	0.00	57.01	41.53	90.91	0.00	0.00	0.00	8.11	29.11	4.71	73.68	59.71	96.19
	LLaVA1.5-7B-Chat	29.52	17.81	86.24	0.19	50.00	0.10	14.56	25.47	10.19	48.53	33.58	87.50	0.00	0.00	0.00	23.73	45.17	16.09	25.86	15.23	85.47	0.41	25.00	0.20	21.30	55.68	13.17
	Qwen2-VL-7B-Instruct	14.07	21.76	10.39	48.67	59.94	40.97	36.99	27.84	55.09	14.96	35.29	9.49	36.66	30.88	45.12	51.21	45.47	58.60	18.57	16.90	20.61	27.35	31.47	24.18	60.70	59.48	61.96
	InternVL2-8B	3.46	14.29	1.97	69.49	57.05	88.88	19.49	30.40	14.34	11.33	33.86	6.80	40.24	25.59	94.11	0.25	50.00	0.12	17.39	12.54	28.38	35.52	38.74	32.79	50.67	59.53	44.11
	Llama-3.2-11B-Vision-Instruct	17.00	17.14	16.85	45.55	53.90	39.45	32.79	27.02	41.70	21.60	34.01	15.82	26.89	26.00	27.85	50.54	43.69	59.95	8.97	12.21	7.09	10.07	32.95	5.94	71.12	60.01	87.26
	InternVL2-26B	1.10	33.33	0.56	31.93	67.48	20.91	42.70	28.39	86.04	1.52	17.86	0.79	40.70	37.35	44.72	55.64	44.97	72.97	0.00	0.00	0.00	14.46	45.16	8.61	73.62	59.84	95.67
	InternVL2-40B	1.06	8.70	0.56	65.04	62.04	68.35	37.01	31.48	44.91	0.31	11.11	0.16	35.65	50.18	27.64	59.22	44.15	89.93	1.26	9.52	0.68	33.88	44.22	27.46	71.39	61.21	85.62
	Qwen2-VL-72B-Instruct	31.35	20.30	68.82	46.98	66.73	36.25	12.77	27.67	8.30	31.69	41.82	25.52	40.76	28.73	70.12	25.41	42.17	18.18	3.17	26.32	1.69	3.17	47.06	1.64	74.33	59.71	98.44
	InternVL2-Llama3-76B	0.56	25.00	0.28	28.72	66.10	18.35	43.00	28.44	88.11	8.40	36.59	4.75	27.12	44.44	19.51	58.35	43.66	87.96	1.82	8.82	1.01	27.52	34.36	22.95	68.01	58.87	80.50
	GLM-4V-Plus	30.76	18.29	96.63	4.56	54.35	2.38	1.85	50.00	0.94	47.38	33.88	78.76	23.66	25.65	21.95	6.03	53.06	3.19	25.86	15.46	79.05	0.40	11.11	0.20	31.53	59.66	21.42
	GPT-4o	11.71	35.21	7.02	10.63	77.92	5.70	42.93	27.82	93.96	5.57	38.00	3.01	8.94	53.33	4.88	58.79	42.38	95.95	5.39	23.68	3.04	0.00	0.00	0.00	74.07	59.57	97.92
	Gemini-1.5 Pro	30.84	25.60	38.76	63.31	64.50	62.17	32.57	38.70	28.11	39.58	35.56	44.62	19.88	40.12	13.21	46.86	42.83	51.72	9.04	41.67	5.07	1.61	50.00	0.82	74.21	59.71	98.01
	Claude-3.5-Sonnet	18.90	28.90	14.04	60.39	64.57	56.71	39.24	31.99	50.75	18.81	32.06	13.31	35.54	38.89	32.72	51.47	42.35	65.60	10.03	22.89	6.42	27.70	42.74	20.49	70.97	60.74	85.34
	o3-mini	26.84	24.82	29.21	59.73	65.98	54.56	39.36	35.75	43.77	45.81	35.41	64.87	25.22	43.94	17.68	38.83	46.56	33.29	25.75	26.94	24.66	29.51	44.26	22.13	69.07	62.54	77.12
	Claude-3-7-Sonnet-Thinking	11.79	21.32	8.15	62.21	63.96	60.55	35.33	29.28	44.53	9.80	34.95	5.70	35.97	36.81	35.16	53.24	42.49	71.25	1.24	7.69	0.68	29.01	46.40	21.11	71.87	60.47	88.56
	GPT-4.1	6.25	42.86	3.37	43.50	72.93	30.99	43.55	29.67	81.89	9.58	43.59	5.38	12.19	51.52	6.91	57.90	42.08	92.75	6.27	43.48	3.38	3.92	45.45	2.05	74.11	59.64	97.83
	Gemini-2.5-flash-preview	30.93	32.51	29.49	39.83	74.48	27.19	42.14	30.14	70.00	22.89	35.31	16.93	21.24	48.20	13.62	55.58	42.91	78.87	9.73	48.48	5.41	11.11	57.69	6.15	74.36	60.33	96.88
LLMs (LoRA)	ChatGLM3-6B-Base (LoRA)	29.72	42.86	22.75	69.34	64.06	75.57	38.73	39.57	37.92	38.47	40.42	36.71	32.37	42.30	26.22	51.04	45.14	58.72	32.60	40.30	27.36	34.20	37.91	31.15	66.91	62.35	72.18
	LLaMA3-8B-Instruct (LoRA)	36.89	46.38	30.62	73.92	70.14	78.14	44.49	44.44	44.53	39.24	41.33	37.34	43.92	52.39	37.80	53.23	48.02	59.71	34.56	40.64	30.07	40.27	45.27	36.27	68.90	64.38	74.09
	InternLM2.5-7B-Chat (LoRA)	38.04	41.70	34.97	34.52	41.67	29.47	54.64	48.30	62.90	38.04	41.70	34.97	34.52	41.67	29.47	54.64	48.30	62.90	31.11	38.69	26.01	35.64	38.48	33.20	66.18	62.06	70.88
	LLaMA3.1-8B-Instruct (LoRA)	14.55	44.29	8.71	68.28	60.19	78.90	35.53	37.01	34.15	34.01	40.57	29.27	34.86	37.56	32.52	51.87	45.93	59.58	20.49	30.07	15.54	35.69	34.21	37.30	63.98	61.45	66.72
	MiniCPM3-4B-Chat (LoRA)	20.39	44.76	13.20	66.70	61.39	73.00	36.87	35.22	38.68	38.29	37.67	38.92	31.26	44.87	23.98	50.00	44.91	56.39	28.82	39.64	22.64	29.48	34.72	25.61	67.03	60.97	74.44
MLLMs (LoRA)	LLaVA1.5-7B-Chat (LoRA)	39.93	54.63	31.46	70.87	64.27	78.99	33.20	36.59	30.38	39.97	39.30	40.66	38.65	45.23	33.74	48.99	46.24	52.09	35.67	43.84	30.07	31.44	33.64	29.51	66.64	62.74	71.06
	LLaVA-1.5-13B-Chat (LoRA)	48.72	56.72	42.70	74.89	70.66	79.66	44.77	46.90	42.83	50.45	52.48	48.58	48.14	50.33	46.14	56.76	53.99	59.83	51.39	56.73	46.96	47.19	47.05	47.34	69.61	68.22	71.06
	LLaVA-NeXT-7B-Chat (LoRA)	42.31	56.02	33.99	70.89	66.39	76.05	35.53	35.98	35.09	40.03	42.73	37.66	35.43	42.69	30.28	49.54	44.33	56.14	45.56	53.15	39.86	41.08	43.21	39.14	69.33	66.12	72.88
	LLaVA-NeXT-13B-Chat (LoRA)	50.23	56.49	45.22	75.75	71.67	80.32	46.81	49.58	44.34	47.51	49.07	46.04	49.68	51.76	47.76	55.95	53.54	58.60	49.91	55.10	45.61	46.84	48.26	45.49	70.47	68.21	72.88
	Qwen2-VL-7B-Instruct (LoRA)	47.91	59.34	40.17	75.37	70.13	81.46	42.79	45.26	40.57	37.50	53.20	28.96	44.57	42.59	46.75	57.60	51.04	66.09	50.83	55.87	46.62	41.87	48.77	36.68	70.62	66.09	75.82
	Qwen2.5-VL-3B-Instruct (LoRA)	53.19	59.58	48.03	76.96	73.86	80.32	46.87	47.93	45.85	50.49	51.83	49.21	49.40	53.70	45.73	57.82	54.52	61.55	50.49	51.83	49.21	49.40	53.70	45.73	57.82	54.52	61.55
	Qwen2.5-VL-7B-Instruct (LoRA)	47.15	55.98	40.73	75.14	72.36	78.14	47.53	46.96	48.11	42.66	49.48	37.50	46.98	49.01	45.12	56.15	50.80	62.78	44.87	51.30	39.86	43.91	46.76	41.39	70.37	67.01	74.09
	Llama-3.2-11B-Vision-Instruct (LoRA)	47.74	67.01	37.08	76.41	73.09	80.04	50.40	47.88	53.21	46.18	49.19	43.51	43.40	51.69	37.40	55.42	49.76	62.53	48.29	65.70	38.18	43.74	53.73	36.89	74.04	66.88	82.93
MLLMs (MCLF)	LLaVA1.5-7B-Chat (MCLF)	56.44	62.16	51.69	78.66	75.30	82.32	52.45	54.47	50.57	50.16	50.98	49.37	53.94	53.61	54.27	57.00	56.52	57.49	56.51	61.85	52.03	55.35	54.58	56.15	72.96	71.95	74.00
	LLaVA-1.5-13B-Chat (MCLF)	56.57	62.08	51.97	78.87	75.86	82.13	53.35	54.89	51.89	52.09	53.99	50.32	55.15	56.50	53.86	57.38	55.23	59.71	59.41	65.45	54.39	56.36	55.96	56.76	74.69	73.35	76.08
	LLaVA-NeXT-7B-Chat (MCLF)	57.78	67.42	50.56	77.79	74.91	80.89	54.43	54.32	54.53	50.58	53.33	48.10	52.47	54.23	50.81	56.66	53.81	59.83	58.49	66.24	52.36	55.83	55.71	55.94	74.02	72.22	75.91
	LLaVA-NeXT-13B-Chat (MCLF)	56.70	62.80	51.69	78.73	76.50	81.08	54.53	54.53	54.53	53.30	52.97	53.64	54.80	57.62	52.24	57.26	55.99	58.60	57.80	60.82	55.07	55.02	55.53	54.51	73.18	72.04	74.35
	Qwen2-VL-7B-Instruct (MCLF)	62.04	65.42	58.99	78.42	77.58	79.28	53.73	53.14	54.34	55.47	56.91	54.11	53.14	58.07	48.98	59.14	55.70	63.02	58.98	59.18	58.78	56.66	55.02	58.40	72.28	73.18	71.40
	Qwen2.5-VL-3B-Instruct (MCLF)	55.66	61.07	51.12	77.38	74.24	80.80	47.80	49.49	46.23	50.48	51.14	49.84	48.43	51.73	45.53	56.14	53.77	58.72	56.75	58.16	55.41	49.95	52.36	47.75	71.29	69.61	73.05
	Qwen2.5-VL-7B-Instruct (MCLF)	58.33	64.73	53.09	78.92	77.47	80.42	56.09	54.87	57.36	55.54	54.77	56.33	51.49	56.12	47.56	58.99	57.06	61.06	58.74	65.29	53.38	54.12	54.34	53.89	73.03	71.29	74.87
	Llama-3.2-11B-Vision-Instruct (MCLF)	53.24	65.31	44.94	78.10	77.23	78.99	53.88	50.08	58.30	56.10	60.81	52.06	56.31	60.00	53.05	63.40	58.52	69.16	60.16	71.30	52.03	55.74	56.81	54.71	75.73	72.76	78.94

Table 5: Experimental results of LLMs, and MLLMs on partial metaphor reasoning tasks.

MLLMs Prompts for Instruction Generation

Metaphor Detection

Metaphor detection task prompt

English: *<image>* The text description for multimodal image data is *<text>* .Please judge whether there is a metaphor according to the input Image-Text Pair. Please provide the answer directly Yes (1) or No (0).

Chinese: <图片>多模态图像数据的文本内容如下： <文本>。根据给定的图文对，请判断是否存在隐喻，直接给出结果‘是’（1）或‘否’（0）。

Source/Target Domain Detection

Source domain detection task prompt

English: *<image>* The text description for multimodal image data is *<text>*. I am doing Vehicle of Metaphor (Source Domain) Detection. Please judge whether the Vehicle of Metaphor exists in the text, image, or text-image complementation according to the input Chinese/English text. Please provide the answer directly text (1), image (2), or text-image complementary (3).

Chinese: <图片>多模态图像数据的中/英文文本内容如下： <文本>。判断隐喻的喻体（源域）是存在于文本、图片、还是图文共存，请直接给出结果‘文本’（1）、‘图片’（2）、‘图文共存’（3）。

Target domain detection task prompt

English: *<image>* The text description for multimodal image data is *<text>*. I am doing Tenor of Metaphor (Target Domain) Detection. Please judge whether the Tenor of Metaphor exists in the text, image, or text-image complementation according to the input Chinese/English text. Please provide the answer directly text (1), image (2), or text-image complementary (3).

Chinese: <图片>多模态图像数据的中/英文文本内容如下： <文本>。判断隐喻的本体（目标域）模态是存在于文本、图片、还是图文共存，请直接给出结果‘文本’（1）、‘图片’（2）、‘图文共存’（3）。

Metaphor Classification

Metaphor classification task prompt

English: *<image>* The text description for multimodal image data is *<text>*. I am doing Metaphor Type Classification. Please judge whether the Metaphor exists in the text, image, or text-image complementation according to the input Chinese/English text. Please provide the answer directly text (1), image (2), or text-image complementary (3).

Chinese: <图片>多模态图像数据的中/英文文本内容如下： <文本>。判断隐喻是存在于文本、图片、还是图文共存，请直接给出结果‘文本’（1）、‘图片’（2）、‘图文共存’（3）。

Tenor/Vehicle Extraction

Tenor extraction task prompt

English: *<image>* Multimodal image data Chinese/English text: *<text>*. What is the metaphor's tenor in the above multimodal data? Answer in Chinese/English.

Chinese: <图片>多模态图像数据中/英文文本为：<文本>。上述多模态数据的隐喻本体是什么？使用中文回答。

Vehicle extraction task prompt

English: *<image>* Multimodal image data Chinese/English text: *<text>*. What is the metaphor's vehicle in the above multimodal data? Answer in Chinese/English.

Chinese: <图片>多模态图像数据中/英文文本为：<文本>。上述多模态数据的隐喻喻体是什么？使用中文回答。

Emotion Analysis

Emotion analysis task prompt

English: *<image>*As Multimodal Image-Text Pairs may have multiple emotions, only the most important emotion should be considered. Please analyze the following image-text pair and determine its emotion category. The options include:

1. happiness: positive sentiments of happiness, relaxation, peace, striving to be strong;
2. love: positive sentiments of desire to explore, yearning, liking, worship, respect, and approval;
3. anger: express negative feelings of anger or irritation;
4. sorrow: including but not limited to sadness, disappointment, grievance, guilt, stress, and despair;
5. fear: negative emotions of panic, fear, avoidance, and apology;
6. hate: negative sentiments of boredom, hatred, demeaning, jealousy, and disgust;
7. surprise: express disbelief, surprise and feelings when things are not expected to happen.

The Chinese/English text content of the Multimodal Image-Text Pair is as follows:*<text>*. Please directly provide the primary emotion type: 'happiness' (1), 'love' (2), 'anger' (3), 'sorrow' (4), 'fear' (5), 'hate' (6), or 'surprise' (7).

Chinese: <图片>由于多模态图文对可能具有多种情感，仅需考虑最重要的情感，请根据如下图文对的内容，判断其最主要的情感类型。选项包括：

1. happiness：表达积极的情感，如快乐、放松、平和、努力变得坚强等；
2. love：表达积极的情感，如探索的欲望、向往、喜欢、崇拜、尊敬和认同等；
3. anger：表达愤怒或恼怒的负面情感等；
4. sorrow：包括但不限于悲伤、失望、委屈、内疚、压力和绝望的情感等；
5. fear：表达恐慌、害怕、回避和道歉的负面情感等；
6. hate：表达厌烦、仇恨、贬低、嫉妒和厌恶的负面情感等；
7. surprise：表达惊讶、不相信和事物发生时的意外感受等。

多模态图文对的中/英文文本内容如下：<文本>。请直接给出表达情感类型的结果：'happiness' (1)、'love' (2)、'anger' (3)、'sorrow' (4)、'fear' (5)、'hate' (6)或'surprise' (7)。

Offensiveness Detection

Offensiveness detection task prompt

English: *<image>*Please determine the Offensive level based on the following Text-Image Pair. The options include:

1. No_Offensive: No intention to express criticism, attack, sarcasm, belittlement, or offense;
2. Slightly: Slightly expresses criticism, attack, sarcasm, belittlement, or offense, which may cause discomfort but not excessively;
3. Moderately: Moderately expresses criticism, attack, sarcasm, belittlement, or offense, which is highly likely to cause discomfort.

The content of the Multimodal Text-Image Pair in Chinese/English is as follows:*<text>*. Please directly provide the Offensive level: 'No_Offensive' (1), 'Slightly' (2), 'Moderately' (3).

Chinese: <图片>请根据如下图文对的内容，判断其Offensive程度。选项包括：

1. No_Offensive：无意表达批判评价、攻击、讽刺、贬低或冒犯；
2. Slightly：轻微表达批判评价、攻击、讽刺、贬低或冒犯，可能会造成不适，但不过度；
3. Moderately：中度表达批判评价、攻击、讽刺、贬低或冒犯，极易造成不适。

多模态图文对的中/英文文本内容如下：<文本>。请直接给出Offensive程度：'No_Offensive'（1）、'Slightly'（2）、'Moderately'（3）。

Intention Recognition

Intention recognition task prompt

English: *<image>*As Multimodal Image-Text Pairs may have multiple intentions, only the most important intention should be considered. Please analyze the following image-text pair and determine its intention category. The options are:

1. interactive: highlight closeness and interaction between the two parties, hoping to get or reduce feedback;
2. expressive: express emotions, social expectations, and image-building;
3. entertaining: mainly for entertainment, leisure, creativity, etc.;
4. offensive: offensive, discriminatory, satirical, insulting, and violent, etc.;
5. other: does not meet any of the above four intentions.

The Chinese/English text content of the Multimodal Image-Text Pair is as follows:*<text>*. Please directly provide the intention type result: 'interactive' (1), 'expressive' (2), 'entertaining' (3), 'offensive' (4), or 'other' (5).

Chinese: <图片>由于多模态图文对可能具有多种意图，仅需考虑最重要的意图，请根据如下图文对的内容，判断其最主要的意图类型。选项包括：

1. interactive：强调双方的亲密关系与互动，希望获取或减少反馈；
2. expressive：表达情感、社会期望，塑造形象等；
3. entertaining：主要用于娱乐、消遣、创意等；
4. offensive：冒犯、歧视、讽刺、侮辱以及暴力等；
5. other：不符合以上四种意图。

多模态图文对的中/英文文本内容如下：<文本>。请直接给出意图类型结果：'interactive'（1）、'expressive'（2）、'entertaining'（3）、'offensive'（4）或'other'（5）。

Humor Detection

Humor detection task prompt

English: *<image>* The text description for Multimodal Image-Text Pair is *<text>*. Please judge whether there is the humor according to the input Image-Text Pair. Please provide the answer directly Yes (1) or No (0).

Chinese: <图片>多模态图文对的中/英文文本内容如下：<文本>。根据给定的图文对，请判断是否存在讽刺，直接给出结果‘是’（1）或‘否’（0）。

Sarcasm Detection

Sarcasm detection task prompt

English: *<image>* The text description for Multimodal Image-Text Pair is *<text>*. Please judge whether there is the sarcasm according to the input Image-Text Pair. Please provide the answer directly Yes (1) or No (0).

Chinese: <图片>多模态图文对的中/英文文本内容如下：<文本>。根据给定的图文对，请判断是否存在幽默，直接给出结果‘是’（1）或‘否’（0）。

Hyperbole Detection

Hyperbole detection task prompt

English: *<image>* The text description for Multimodal Image-Text Pair is *<text>*. Please judge whether there is the hyperbole according to the input Image-Text Pair. Please provide the answer directly Yes (1) or No (0).

Chinese: <图片>多模态图文对的中/英文文本内容如下：<文本>。根据给定的图文对，请判断是否存在夸张，直接给出结果‘是’（1）或‘否’（0）。